25900515 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEMATA02 - NUMERICAL METHODS FOR SCIENTIFIC COMPUTING

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Convert the hexadecimal number $2F_{16}$ into decimal. [U] / [CO1]

2. Define numerically unstable algorithm. [U] / [CO1]

3. Perform one iteration of the bisection method for $f(x) = x^2 - 10$ on $[3, 4]$. [A] / [CO2]

4. What is meant by forward elimination in Gaussian elimination? [K] / [CO2]

5. Write the LU factorization of $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$. [K] / [CO2]

6. What is meant by closed-type integration rules? [K] / [CO4]

7. Write the steps of Richardson extrapolation used to improve the solution obtained by Euler method. [U] / [CO5]

8. State the theorem for existence and uniqueness of solutions of an initial value problem. [K] / [CO5]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **6** marks

9. Convert $(0.859375)_{10}$ and $(0.7)_{10}$ to the corresponding binary fraction. [U] / [CO1]

10. Derive Gauss–Seidel iteration equations for [A] / [CO2]
$2x + y = 4, \quad x + 3y = 5$
and compute the first iteration.

11. Write the Jacobi iteration in matrix form and perform 2 iterations for [A] / [CO2]
$4x + y = 7, \quad x + 3y = 8$ with initial guess $(0, 0)$

12. Find the natural cubic spline that fits the data [K] / [CO3]

x	1	2	3	4
$f(x)$	0	1	0	0

13. Using least squares, fit a straight line to the data: $(0, 0.2)$, $(0.2, 0.447)$, $(0.4, 0.632)$, $(0.6, 0.775)$, $(0.8, 0.894)$. [A] / [CO3]

14. Using the data $f(0) = 0, f(1) = 1, f(2) = 8$, approximate $f(1.5)$ using piecewise linear interpolation. [A] / [CO3]

15. Derive the error term in interpolation-based numerical integration. [U] / [CO4]

16. Derive the expression for the first derivative of the interpolation error $E_n'(x)$. [U] / [CO4]

[6x5 = 30]

Part C

Essay Questions

Answer any **3** questions

Each question carries **10** marks

17. Define a normalized floating-point number. Represent -2735.46875 in normalized floating-point form (base 2 and base 10). [K] / [CO1]

18. Find the root of $f(x) = x^3 - 2x^2 - 5$ correct up to 4 decimal places using Newton–Raphson method with $x_0 = 2$. [A] / [CO2]

19. Solve the system of equations using matrix methods: [A] / [CO2]
$x + 2y + 3z = 1, \quad 2x + 5y + 8z = 3, \quad 3x + 8y + 14z = 4.$

20. [A] / [CO3]
    1. Obtain the Lagrange linear interpolating polynomial for $f(x) = \sin x$ on $[1, 3]$ and estimate the bound on the truncation error. Approximate $f(1.5)$ and $f(2.5)$.
    2. Repeat by dividing the interval into subintervals $[1, 2]$ and $[2, 3]$ and compare the errors.

21. Apply the fourth-order Runge-Kutta method to approximate $y(0.4)$ for the initial value problem
$y' = x^2 + y^2, y(0) = 0$, using $h = 0.2$. [A] / [CO5]

22. Compute an approximation to $u(1), u'(1)$ and $u''(1)$ with Taylor series method of second order
and step length
$h = 1,$
for the initial value problem $u''' + 2u'' + u' - u = cost$, where t is in $[0, 1]$,
$u(0) = 0, u'(0) = 1, u''(0) = 2$
after reducing it to a system of first order equations. [An] / [CO5]

[10x3 = 30]